(a)

The panel data model: $$y_{it} = \beta_0 + \beta_1 x_{it} + a_i + v_{it}$$

$$t = 1: \quad y_{i1} = \beta_0 + \beta_1 x_{i1} + a_i + v_{i1}$$
$$t = 2: \quad y_{i2} = \beta_0 + \beta_1 x_{i2} + a_i + v_{i2}$$

Subtract the first equation from the second equation and obtain

$$y_{i2} - y_{i1} = \beta_1(x_{i2} - x_{i1}) + (v_{i2} - v_{i1})$$
$$\Longrightarrow \quad \Delta y_i \quad = \quad \beta_1 \Delta x_i + \Delta v_i, \text{ where } i = 1,2 \ldots, n$$

(b) First differenced equation: $\Delta y_i = \alpha_0 + \ \beta_1 \Delta x_i + \Delta v_i$, where $i = 1,2 \ldots, n$

$\hat{\beta}_{1,FD} = \frac{\sum_{i=1}^{n}(\Delta x_i - \overline{\Delta x})(\Delta y_i - \overline{\Delta y})}{\sum_{i=1}^{n}(\Delta x_i - \overline{\Delta x})^2}$ and $\hat{\alpha}_{0,FD} = \overline{\Delta y} - \hat{\beta}_{1,FD}\overline{\Delta x}$

(c)

(1) Linear in parameters; (2) Random Sampling; (3) No perfect collinearity

(4) Zero conditional mean assumption $E(\Delta v_i | \Delta x_i) = E(\Delta v_i) = 0$

If we write Δx_i as x_i^* and Δy_i as y_i^*, then it's clear that

$\hat{\beta}_{1,FD} = \frac{\sum_{i=1}^{n}(x_i^* - \overline{x^*})(y_i^*, - \overline{y^*})}{\sum_{i=1}^{n}(x_i^* - \overline{x^*})^2}$ is the "OLS" estimator. So, this estimator is consistent and unbiased.

$$\begin{aligned}
\hat{\beta}_{1,FD} &= \frac{\sum_{i=1}^{n}(\Delta x_i - \overline{\Delta x})(\Delta y_i - \overline{\Delta y})}{\sum_{i=1}^{n}(\Delta x_i - \overline{\Delta x})^2} \ = \frac{\sum_{i=1}^{n}(\Delta x_i - \overline{\Delta x})(\alpha_0 + \ \beta_1 \Delta x_i + \Delta v_i)}{\sum_{i=1}^{n}(\Delta x_i - \overline{\Delta x})^2} \\
&= \alpha_0 \frac{\sum_{i=1}^{n}(\Delta x_i - \overline{\Delta x})}{\sum_{i=1}^{n}(\Delta x_i - \overline{\Delta x})^2} + \ \beta_1 \frac{\sum_{i=1}^{n}(\Delta x_i - \overline{\Delta x})\Delta x_i}{\sum_{i=1}^{n}(\Delta x_i - \overline{\Delta x})^2} + \frac{\sum_{i=1}^{n}(\Delta x_i - \overline{\Delta x})\Delta v_i}{\sum_{i=1}^{n}(\Delta x_i - \overline{\Delta x})^2} \\
&= \ \beta_1 + \frac{\sum_{i=1}^{n}(\Delta x_i - \overline{\Delta x})\Delta v_i}{\sum_{i=1}^{n}(\Delta x_i - \overline{\Delta x})^2} \approx \ \beta_1 + \frac{Cov(\Delta x_i, \Delta v_i)}{Var(\Delta x_i)}, \ \ when\ n\ goes\ to\ infitniy \\
&= \ \beta_1
\end{aligned}$$

(d) $Var(\Delta v_i | \Delta x_i) = \sigma^2_{\Delta v_i}$

The serial correlation is not an issue here because the original data only has two periods (t=1,2). Once we take the first difference, no any time periods "t" found anymore in the transformed data.$\Delta y_i = \alpha_0 + \ \beta_1 \Delta x_i + \Delta v_i$, where $i = 1,2 \ldots, n$

(2)

(a) $\log(rent) = \beta_0 + \gamma dummy_{1990} + \beta_1 \log(avgin) + \beta_2 \log(pop) + \beta_3 pctstu + u$

```
. reg lrent y90 lavginc lpop pctstu

      Source |       SS           df       MS      Number of obs   =       128
-------------+----------------------------------   F(4, 123)       =    190.92
       Model |  12.1080112         4  3.02700281   Prob > F        =    0.0000
    Residual |   1.9501234       123  .015854662   R-squared       =    0.8613
-------------+----------------------------------   Adj R-squared   =    0.8568
       Total |  14.0581346       127  .110693974   Root MSE        =    .12592

------------------------------------------------------------------------------
       lrent |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
         y90 |   .2622267   .0347632     7.54   0.000     .1934151    .3310384
     lavginc |   .5714461   .0530981    10.76   0.000     .4663417    .6765504
        lpop |   .0406863   .0225154     1.81   0.073    -.0038815    .0852541
      pctstu |   .0050436   .0010192     4.95   0.000     .0030262     .007061
       _cons |  -.5688069   .5348808    -1.06   0.290    -1.627571    .4899568
------------------------------------------------------------------------------
```

Other things being equal, on average the rents grew by about 26% compared to year 1980.
Other things being equal, one percentage point increased in *pctstu* increases *rent* by 0.5% on average.

(b) $\Delta\log(rent) = \alpha_0 + \beta_1 \Delta \log(avgin) + \beta_2 \Delta \log(pop) + \beta_3 \Delta pctstu + \Delta u$

```
. reg clrent clavginc clpop cpctstu

      Source |       SS           df       MS      Number of obs   =        64
-------------+----------------------------------   F(3, 60)        =      9.51
       Model |  .231738668         3  .077246223   Prob > F        =    0.0000
    Residual |  .487362198        60  .008122703   R-squared       =    0.3223
-------------+----------------------------------   Adj R-squared   =    0.2884
       Total |  .719100867        63  .011414299   Root MSE        =    .09013

------------------------------------------------------------------------------
      clrent |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
    clavginc |   .3099605   .0664771     4.66   0.000     .1769865    .4429346
       clpop |   .0722456   .0883426     0.82   0.417     -.104466    .2489571
     cpctstu |   .0112033   .0041319     2.71   0.009     .0029382    .0194684
       _cons |   .3855214   .0368245    10.47   0.000     .3118615    .4591813
------------------------------------------------------------------------------
```

Other things being equal, one percentage point increase in *pctstu* increased rents by about 1.1%. Not surprisingly, we obtain a much less precise estimate when we difference, i.e., standard errors are bigger. While we have differenced away a_i, there may be other unobservable variables that change over time and are correlated with Δ*pctstu*.

(c)

```
. reg clrent clavginc clpop cpctstu, robust

Linear regression                                      Number of obs     =         64
                                                       F(3, 60)          =      11.30
                                                       Prob > F          =     0.0000
                                                       R-squared         =     0.3223
                                                       Root MSE          =     .09013
```

clrent	Coef.	Robust Std. Err.	t	P>|t|	[95% Conf.	Interval]
clavginc	.3099605	.0893099	3.47	0.001	.1313141	.488607
clpop	.0722456	.0696796	1.04	0.304	-.0671344	.2116255
cpctstu	.0112033	.002936	3.82	0.000	.0053305	.0170762
_cons	.3855214	.0487186	7.91	0.000	.2880697	.4829731

The heteroskedasticity-robust standard error on Δ*pctstu* is about .0028, which is much smaller than the usual OLS standard error. This only makes *pctstu* even more significant (robust *t* statistic $\approx$ 4).